Multivariate Calculus

SCIENCE PROGRAM
COURSE OUTLINE
WINTER 2026

General Information.
Discipline: Mathematics *Course code*: 201-SNB-AB
Ponderation: 2-2-2 *Credits*: 2 *Prerequisite*: 201-SN3-RE
Objective: Consolidate one's scientific culture in a scientific field (0GNF)

Students are strongly advised to seek help from their instructor as soon as they encounter difficulties in the course.

Introduction. Multivariate Calculus is the last course in the Calculus sequence. It is offered as an option course in the Science Program for students intending to pursue university studies primarily in Engineering, Physics, Mathematics or Chemistry. Multivariate Calculus extends the basic concepts of Differential Calculus and Integral Calculus vector-valued functions and to functions of several variables. Following a short introduction to vector geometry the course dives into vector functions, their derivatives and integrals. This course covers functions of several variables with emphasis on partial derivatives, directional derivatives and gradient vectors, vector fields and potential functions. The course progresses with line integrals, double integrals and change of variables in double integrals (mainly polar coordinates). Finally, the course covers Green's theorem, the fundamental theorem of calculus for double integrals, concluding with applications of Green's theorem to the curl and divergence of a vector field. Emphasis will be placed on clarity and rigour in reasoning and in the application of methods. Some of the applied problems solved in this course will be taken from other disciplines in the program, primarily Physics.

Teaching Methods. This course will be 60 hours, meeting twice per week for a total of four hours per week. The course will rely mainly on the lecture method. The following methods may also be used: question-and-answer sessions, computer lab sessions, problem-solving sessions, class discussions, and assigned reading for independent study.

The use of cell phones, laptops, or similar technology for any purpose that is not directly related to the course is not permitted.

Textbook. Your teacher may require the book *Multivariable Calculus, by James Stewart* (8th or 9th Edition). It is available from the bookstore for about $170.
An additional source is available, free of charge: *Vector Calculus* by *Michael Corral*: https://www.mecmath.net/

Course Costs. In addition to the cost of the textbook (see above), your instructor might recommend you acquire an inexpensive scientific calculator ($15-$20).

Other Resources.

Evaluation. The *Final Grade* will be the better of:

- 50% Final Exam plus 50% Class Mark
- 75% Final Exam plus 25% Class Mark

The Final Evaluation in this course consists of the Final Exam, which covers all elements of the competency. The Class Mark consist of results from two or more in-class written tests (worth 60% - 75% of the Class Mark, where at least 75% of the Class Mark is composed of tests or quizzes), and minor assessments (worth 25% of the Class Mark). The specifics of the Class Mark will be given by each instructor during the first week of classes in an appendix to this outline. The Final Exam is set by the Course Committee (which consists of all instructors currently teaching this course), is written during the examination period, and is marked by each individual instructor.

A student whose Class Mark is less than 50% may choose not to take the Final Exam in which case their grade will be their Class Mark or 50%, whichever is less.

Calculators are not permitted during quizzes, tests or the final exam.

Students must be available until the end of the final examination period to write exams.

Note that in the event of unexpected changes to the academic calendar, the evaluation plan may be modified.

College Policies.
Policy No. 7 - IPESA, Institutional Policy on the Evaluation of Student Achievement: https://www.johnabbott.qc.ca/wp-content/uploads/2021/05/Policy-No.-7-IPESA-FINAL.pdf.

Changes to Evaluation Plan in Course Outline (Article 5.3). Changes require documented unanimous consent from regularly attending students and approval by the department and the program dean.

Religious Holidays (Article 3.2.13 and 4.1.6). Students who wish to miss classes in order to observe religious holidays must inform their teacher of their intent in writing within the first two weeks of the semester.

Student Rights and Responsibilities: (Article 3.2.18). It is the responsibility of students to keep all assessed material returned to them and/or all digital work submitted to the teacher in the event of a grade review. (The deadline for a Grade Review is 4 weeks after the start of the next regular semester.)

Student Rights and Responsibilities: (Article 3.3.6). Students have the right to receive graded evaluations, for regular day division courses, within two weeks after the due date or exam/test date, except in extenuating circumstances. A maximum of three (3) weeks may apply in certain circumstances (ex. major essays) if approved by the department and stated on the course outline. For evaluations at the end of the semester/course, the results must be given to the student by the grade submission deadline (see current Academic Calendar). For intensive courses (i.e.: intersession, abridged courses) and AEC courses, timely feedback must be adjusted accordingly.

Academic Procedure: Academic Integrity, Cheating and Plagiarism (Article 9.1 and 9.2). Cheating and plagiarism are unacceptable at John Abbott College. They represent infractions against academic integrity. Students are expected to conduct themselves accordingly and must be responsible for all of their actions. For more information, see the Academic Procedure on Academic Integrity.

College definition of Cheating: Cheating means any dishonest or deceptive practice relative to examinations, tests, quizzes, lab assignments, research papers or other forms of evaluation tasks. Cheating includes, but is not restricted to, making use of or being in possession of unauthorized material or devices and/or obtaining or providing unauthorized assistance in writing examinations, papers or any other evaluation task and submitting the same work in more than one course without the teacher's permission. It is incumbent upon the department through the teacher to ensure students are forewarned about unauthorized material, devices or practices that are not permitted.

College definition of Plagiarism: Plagiarism is a form of cheating. It includes copying or paraphrasing (expressing the ideas of someone else in one's own words), of another person's work or the use of another person's work or ideas without acknowledgement of its source. Plagiarism can be from any source including books, magazines, electronic or photographic media or another student's paper or work.

Policy No. 13 - Policy on Student Conduct and Discipline Procedures: https://www.johnabbott.qc.ca/wp-content/uploads/2021/09/POLICY-13_-_policy_on_student_conduct-Final-BOG-2021-09-21.pdf.

Course Content (with selected exercises). The exercises listed below (referring to the 8th edition of Stewart) should help you practice and learn the material taught in this course. Your teacher may supplement this list during the semester. Regular work done as the course progresses should make it easier for you to master the course.

Vector geometry review.

12.2 (vectors): 6, 7, 22, 44, 45
12.3 (dot product): 5, 6, 15, 17, 23, 26, 31, 41, 55, 56, 63, 64
12.4 (cross Product): 1, 7, 13, 15, 16, 29, 35, 44-46, 50-53
12.5 (lines and planes): 1, 4, 5, 9, 12, 13, 19, 20, 26, 27, 30, 33, 35, 37, 39, 40, 45, 53, 55, 57, 61, 64, 71, 73, 76, 79, 80

Parametrized curves.

13.1 (parametrized curves): 19, 20, 27, 42, 43, 50
13.2 (derivatives): 9-12, 16-18, 23-28, 35-40, 47-57
13.3 (arc length): 1-6, 13-15

Functions of several variables.

14.1 (multivariate functions): 13, 15–19, 21, 22, 45–52
14.3 (partial derivatives): 15-44, 51-70, 76-80
14.5 (chain rule): 1-13, 15, 16, 21-26, 38, 45-54
14.6 (directional derivatives and gradient): 5, 9, 12, 15, 16, 22, 24, 25, 28, 29, 41, 43, 45, 46, 54, 59, 60, 63, 67, 69

Potential functions and line integrals.

16.1 (vector fields): 11-14, 21, 23, 24, 29–32, 33
16.2 (path integrals): 2, 5, 7, 8, 9, 13–17, 19–22, 33, 39, 43, 50
16.3 (dependence on path): 3, 5, 7, 10, 11, 13–15, 17, 18, 20, 24, 28–30, 32–34

Double integrals.

15.1 (double and repeated integrals): 10, 16, 19, 20, 22, 24, 27, 29, 30, 33, 34, 37, 41, 44, 49
15.2 (double and repeated integrals): 1-25, 31-42, 44
15.3 (polar coordinates): 3, 6, 7, 9, 14, 16, 17, 20, 22, 24, 27, 29, 31, 32, 35, 52, 53, 56, 58, 66, 68
15.9 (change of variables): 2, 3, 5, 15, 16, 19, 20, 23, 24, 25, 26, 27, 28
15.4 (moments and centres of mass): 3, 4, 5, 9, 17 (time permitting)

Green's theorem.

16.4 (Green's theorem): 2, 3, 6, 7, 10, 12, 13, 14, 18, 20–23, 27, 28, 29
16.5 (curl and divergence): 3, 4, 5, 7, 9–11, 13, 15, 17, 18, 21, 22, 25–29, 31, 33–36

Time permitting.

14.7 extreme values (time permitting)
14.8 Lagrange multipliers (time permitting)
Differentiating under the integral sign: instructor's notes and exercises

OBJECTIVES	STANDARDS
Specific Performance Criteria	**Intermediate Learning Objectives ("A student is expected to ...")**
1 *Parametrized curves*	
1.1 Calculus of parametrized curves	1.1.1 Compute and manipulate parametrizations of curves. 1.1.2 Compute and manipulate derivatives and integrals of parametrized curves.
1.2 Arc length	1.2.1 Compute and manipulate speeds and lengths of paramatrized curves.
2 *Functions of several variables*	
2.1 Graphs and level sets	2.1.1 Compute the domain and sketch level sets of a multivariate function.
2.2 Partial derivatives	2.2.1 Compute and manipulate partial derivatives, differentials, directional derivatives and gradients (including use of the chain rule). 2.2.2 State conditions for differentiability and equality of mixed partials. 2.2.3 Verify that a multivariate function satisfies a partial differential equation. 2.2.4 Compute and apply representations of tangent planes and normal lines.
2.3 Differentiating under the integral	2.3.1 State conditions which justify differentiating under the integral, and apply the technique.
3 *Line integrals*	
3.1 Vector fields	3.1.1 State and apply the definitions of a vector field and potential function. 3.1.2 Compute and manipulate the gradient field of a multivariate function. 3.1.3 State and apply the basic condition for a vector field to be conservative. 3.1.4 State and prove the law of conservation of energy.
3.2 Line integrals	3.2.1 Compute and apply line integrals (of scalar functions and vector fields). 3.2.2 State and apply the theorem on path independence of line integrals of vector fields.
4 *Double integrals*	
4.1 Double integrals	4.1.1 Define double integrals and state their basic properties. 4.1.2 Evaluate double integrals via repeated integrals using Fubini's theorem. 4.1.3 Evaluate double integrals by change of variables (esp. polar coordinates). 4.1.4 Evaluate moments and centres of mass using double integrals.
5 *Green's theorem*	
5.1 Green's theorem	5.1.1 State and prove simple cases of Green's theorem. 5.1.2 Use Green's theorem to evaluate line integrals and double integrals. 5.1.3 Use Green's theorem to characterize conservative vector fields.
5.2 Curl and divergence	5.2.1 Define the and compute curl and divergence of a vector field. 5.2.2 Prove a selection of properties of curl and divergence.